

Raphael Bernardo

Senior Android Engineer • *Ex-Uber*
Architect of a streaming framework powering
50+ TV brands — native (JNI/C++), TV & multi-
DRM expertise.

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§ 01 Summary

PROFILE • POSITIONING

Senior Android engineer with cross-stack reach — Uber Maps SDK (JNI/C++), OTT / CTV streaming frameworks, broadcast-grade DRM. Ten-plus years shipping production apps across Android, Android TV, Fire OS, webOS and Tizen — including an OTT streaming framework powering 50+ broadcast brands and 4K multi-DRM playback. As an independent contractor I run multiple client engagements in parallel — at any given time only one is full-time, with the rest as part-time concurrent contracts.

§ 02 Experience

ENGAGEMENTS • 2014 → PRESENT

Nexstar Media Group, Inc.

Aug 2025 — Present • current

Senior Fire OS / Android TV Engineer • via Nimble.LA • full remote

FULL-TIME CONTRACT

- Architected an enterprise **OTT streaming framework** powering **50+ TV station brands** across Fire TV and Android TV using **MVVM + Clean Architecture** with a plugin-based integration layer for third-party services.
- Built five specialised **Media3 / ExoPlayer** implementations covering VOD, live, and audio playback with HLS adaptive bitrate.
- Shipped DRM-protected playback (**Widevine, PlayReady, FairPlay**) with both server-side and client-side ad insertion.
- Integrated **12+ analytics platforms** (Adobe Analytics, mParticle, Nielsen DCR, ComScore, Permutive, BlueConic) under GDPR / CCPA compliance.
- Delivered TV-optimised auth (passwordless, magic link, QR-code), entitlement & flexible paywall, and zero-downtime feature rollouts across **50+ product flavors** via remote config; built Leanback TV UI with full d-pad navigation.
- Refactored callback chains to reactive Kotlin **SharedFlow**; resolved fragment-lifecycle leaks; wired **Firebase Crashlytics + New Relic** with JUnit, Mockk, Robolectric, Espresso coverage.

Kotlin

Coroutines / Flow

Media3 / ExoPlayer

Dagger 2

Retrofit2

Moshi

Glide

Android Leanback

Jetpack (Nav, ViewModel, Room)

Firebase

Google IMA

Adobe Marketing SDK

mParticle

Nielsen

ComScore

New Relic

Mux Analytics

JUnit • Mockk • Espresso

FreeCast, Inc.

Dec 2024 — Jul 2025 · 8 mo

Android / Android TV / Fire OS Engineer · full remote CONTRACT

- Returned to the Select TV and FreeCast Watch products across Android, Android TV, Fire OS, webOS and Tizen through to the end of the engagement.

Claro TV+

Apr 2024 — Dec 2024 · 9 mo

Android TV / Fire OS Engineer · full remote CONTRACT

- Greenfield Android TV / Fire OS app for one of Brazil's largest pay-TV operators; **4K streaming** via DASH / HLS with multi-DRM.
- Architected the data layer as a shared **Kotlin Multiplatform (KMP)** submodule consumed by Android, webOS and Tizen clients.
- Multi-module **MVVM** architecture; Media3 playback; analytics across Firebase + Google Analytics 4.

Kotlin Multiplatform (KMP)

Kotlin

Jetpack Navigation · ViewModel · Room

Hilt

Media3

Coroutines / Flow

Firebase + GA4

webOS · Tizen

FreeCast, Inc.

May 2022 — Apr 2024 · 1 yr 11 mo

Android / Android TV / Fire OS Engineer · full remote CONTRACT

- Led the **full migration of the legacy Java codebase to Kotlin**; introduced Use-Case + Clean Architecture and improved end-user UX.
- Implemented **HLS / DASH** playback for **live linear TV and VOD** with Widevine DRM-protected streaming; integrated Google Cast and a paid subscription gateway.
- Maintained and developed companion apps on **webOS and Tizen** alongside the Android TV / Fire OS surfaces.
- Integrated **OTA tuner platforms** (HDHomeRun, AirTV, Alticast, PDAQ) with GStreamer-based media pipelines for secure live broadcast over HLS / DASH.
- Owned **DRM, EPG, tuner orchestration, transcoding**, and multi-device streaming end-to-end.
- Shipped on Google Play (*Select TV Android, FreeCast Watch*) and Amazon (*FreeCast Watch Fire OS*).

Kotlin

Jetpack Compose

Hilt

Coroutines / Flow

Media3

Room · DataStore

Retrofit2

Picasso

Firebase Analytics · Crashlytics · FCM

Mixpanel

webOS · Tizen

HDHomeRun · AirTV · Alticast · PDAQ

GStreamer

Widevine DRM

Uber Technologies

May 2021 — May 2022 · 1 yr

Android Engineer · Maps SDK team · via Matchpoint Solutions · full remote CONTRACT

- Worked on the **Maps team SDK** module — JNI/C++ bindings between Uber's native Maps engine and the Android client.
- Contributed to Uber's open-source Android architecture framework (**RIBs**); shipped to Uber Rider, Uber Eats and Uber Driver.
- Operated inside a 10K+ engineer codebase using **Buck, Flipper**, custom analytics / logger / crash reporter and a feature-by-module architecture.

Java · Kotlin

C++ · NDK · JNI

Groovy

Python · SQL

RIBs

Buck

CMake

Flipper

Firebase

FreeCast, Inc.

Mar 2021 — May 2021 · 3 mo

Android / Android TV / Fire OS Engineer · full remote CONTRACT

- Opening engagement on the Select TV Android and FreeCast Watch codebases, starting the migration from Java to Kotlin.

Super Revendedores

Oct 2018 – Feb 2021 · 2 yrs 5 mo

Android Engineer · full remote

- Migrated the full codebase from Java to Kotlin; implemented **online payment gateway**, virtual showcases and an SSO link to a partner web ordering portal.
- Built scalable asynchronous order processing on **AWS SQS / SNS** for reliable queuing and notification delivery.

Kotlin

AWS (SQS · SNS · Storage)

Firebase (Firestore · Auth · Remote Config · IAM · FCM)

Facebook OAuth

DbFlow

Dagger 2

MVP

Retrofit2

Concrete · an Accenture Company

Dec 2016 – Oct 2018 · 1 yr 11 mo

Jr. Android Engineer

- Built the **Minha CVC** travel app from scratch — **1M+ downloads**; shop geolocation directing customers to physical stores; itinerary feature reducing customer-service call volume.
- Worked on **Getnet**; migrated MVP to MVVM; introduced RxJava2 and LiveData.

Java 8

Kotlin

RxJava2 · LiveData

MVP → MVVM

Dagger 2

Google Maps API

Retrofit2 · Picasso

Firebase

Jenkins CI/CD

PraPagar (fintech)

Jun 2015 – Dec 2016 · 1 yr 7 mo

Jr. Android Engineer · collaborating with US team (MaxxPotential)

- Built mobile payments via **QR code and geolocation**; first Android + Laravel REST experience.

Android

PHP · Laravel 5

REST

OOP · Design Patterns

FASOLTI · Fábrica de Soluções em TI

Sep 2014 – Jan 2015 · 5 mo

Test Developer · INTERNSHIP

- Documented and implemented automated test cases in **Selenium with TestNG**.

IBM Brasil

Jun 2014 – Dec 2014 · 6 mo

Software Engineering Intern · Rio de Janeiro · while pursuing B.S. at Instituto Infnet

- Wrote **SQL queries against IBM DB2** in support of contract and sales teams; produced reporting spreadsheets and Hyperion-based BI artefacts.
- Documented internal processes via SharePoint; first professional exposure to enterprise databases, REST APIs, and Agile workflows.

IBM DB2

SQL

IBM Hyperion

SharePoint

Python

REST APIs

Git

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§ 03 Selected Personal Work

INDEPENDENT · OPEN SOURCE · RESEARCH

Arquive

Self-hosted personal media archive & streaming server. In-browser **HLS playback** with GPU-accelerated transcoding cache, **IPTV** (M3U + XMLTV EPG + recording), DLNA/UPnP for smart TVs, distributed GPU-fleet transcoding, FAISS-backed face recognition, local-AI captions via Ollama, plus a native **Apple TV (SwiftUI)** client.

PythonReactHLS · GPU Transcoding

FAISS · InsightFaceOllama · Qwen2.5-VLDLNA

M3U · XMLTVSwiftUI tvOS

Socials Automator

End-to-end Instagram carousel generator — AI writes the captions, AI generates the images, then auto-posts. Live at [@ai.for.mortals](#). Runs on **100% local AI** (LM Studio + ComfyUI) at zero cost, with cloud fallback (Z.AI, OpenAI, Groq, Gemini, fal.ai). Smart slide-count selection, post-history awareness, scheduled loop mode.

PythonLM StudioComfyUI · Stable Diffusion

Z.AI · OpenAI · Groq · Gemini

Instagram Graph APICloudinary

DynamicMusicApp

Android music player with **three pluggable design systems** users switch at runtime — *Material You* (Google), *Neon Wave* (synthwave), and *Organic Flow* (botanical). One Compose codebase, three radically different visual identities — design-system theming treated as a first-class architectural concern.

KotlinJetpack ComposeMaterial 3MVI

Coroutines / FlowHilt

Infinite Research

Autonomous AI research system that refines documents through infinite self-learning iterations. **Agno agents** orchestrate **1–10 parallel DuckDuckGo searches** per iteration; each pass evolves search terms and patches identified gaps. Local LLMs via **LMStudio**, vector-DB RAG over prior refinements, SQLite versioning, graceful shutdown.

PythonAgno AgentsLMStudio (local LLM)

RAG · Vector DBDuckDuckGo SearchSQLite

§ 04 Education

FORMAL · CONTINUING

Instituto Infnet · Rio de Janeiro

2013 – 2018

B.S. Computer Engineering — software development, mathematical modelling, OS & hardware architecture.

Chiswick House School (Malta)

1997 – 2005

K-12 / High School Diploma.

Orion Consulting · Rio de Janeiro

2010

Java fundamentals — OOP, software design, code patterns and logic.

§ 05 Technical Stack

REFERENCE · ATS INDEX

LANGUAGES

KotlinJavaC++TypeScriptJavaScriptPythonGroovySQL

MOBILE / TV

- PHP (Laravel)
- Android SDK
- Android TV
- Fire OS
- webOS
- Tizen
- Android Leanback
- Jetpack Compose
- Compose for TV
- Kotlin Multiplatform (KMP)
- NDK · JNI

STREAMING · MEDIA

- OTT
- CTV (Connected TV)
- Linear TV
- VOD
- Live Streaming
- Media3 / ExoPlayer
- MediaCodec
- HLS
- LL-HLS
- DASH
- CMAF
- H.264 / AVC
- HEVC / H.265
- HDR10+
- AAC
- Hardware-accelerated decoding
- Widevine
- PlayReady
- FairPlay
- Google IMA SDK
- SSAI · CSAI
- GStreamer
- HDHomeRun · AirTV · Alticast · PDAQ

ARCHITECTURE

- MVVM
- MVI
- Unidirectional Data Flow
- Clean Architecture
- Use Case
- Multi-module
- RIBs
- MVP
- Plugin-based integration

JETPACK · REACTIVE

- Coroutines · Flow
- SharedFlow · StateFlow
- LiveData
- ViewModel · Navigation
- Room · DataStore
- Hilt · Dagger 2
- RxJava2

CLOUD · BACKEND

- Firebase (Analytics · Crashlytics · FCM · Auth · Firestore · Remote Config · IAM)
- AWS (SQS · SNS · Storage)
- Supabase
- PostgreSQL
- REST APIs
- Retrofit2 · Moshi

ANALYTICS

- Adobe Analytics
- Adobe Marketing SDK
- mParticle
- Nielsen DCR
- ComScore
- Permutive
- BlueConic
- New Relic
- Mux
- Mixpanel
- Google Analytics 4

TESTING · BUILD

- JUnit
- Mockk
- Robolectric
- Espresso
- Paparazzi (screenshot)
- Turbine (Flow testing)
- Selenium · TestNG
- Buck
- Gradle
- CMake
- Jenkins CI/CD
- Flipper

PERFORMANCE · OBSERVABILITY

- Quality of Experience (QoE)
- Time-to-First-Frame (TTFF)
- Rebuffering rate
- Startup-time optimisation
- Systrace
- Perfetto
- Android Profiler

COMPLIANCE · ACCESSIBILITY

- GDPR
- CCPA
- WCAG 2.2
- TalkBack · d-pad focus order
- DRM & Content Security
- OAuth · SSO

LANGUAGES (SPOKEN)

- Portuguese (native)
- English (fluent)